$$x(x+1)(x+2)(x+3) = a \quad , \quad a \in \mathbb{R} \, , \; x \in \mathbb{R}$$

$\hookrightarrow P(x) \Rightarrow$ polim. grau 4.

$$P(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4$$

$$(x+\alpha)(x+\beta) = x^2 + (\alpha+\beta)x + \alpha\beta$$

$$x(x+3) = \boxed{x^2 + 3x}$$

$$(x+1)(x+2) = \boxed{x^2 + 3x} + 2 \quad , \quad y = x^2 + 3x$$

$$x(x+3)(x+1)(x+2) = a$$

$$\underbrace{(x^2+3x)}\left(\underbrace{x^2+3x}+2\right) = a$$

$$y(y+2) = a$$

$$y^2 + 2y - a = 0$$

$$\boxed{\begin{array}{l} y^2 + py + q = 0 \\ y = -\dfrac{p}{2} \pm \dfrac{1}{2}\sqrt{p^2 - 4q} \end{array}}$$

FORM. DE BHASKARA

$$y = -1 \pm \frac{1}{2}\sqrt{4 + 4a}$$

$$\boxed{y = -1 \pm \sqrt{1+a}}$$

$$y = x^2 + 3x = -1 \pm \sqrt{1+a} \quad \therefore$$

$$b = -1 + \sqrt{1+a}$$

$$c = -1 - \sqrt{1+a}$$

I) $x^2 + 3x = b \;\; \begin{matrix} \nearrow x_1 \\ \searrow x_2 \end{matrix}$

II) $x^2 + 3x = c \;\; \begin{matrix} \nearrow x_3 \\ \searrow \end{matrix}$

$$\{x_1, x_2, x_3, x_4\}$$

ℓ_4

$$x(x+1)(x+2)(x+3)(x+4)(x+5)=a$$

$$x(x+5)=x^2+5x$$

$$(x+2)(x+3)=x^2+5x+6$$

$$(x+1)(x+4)=x^2+5x+4$$

$$y=x^2+5x$$

$$x(x+5)\,(x+2)(x+3)\,(x+1)(x+4)=a$$

$$(x^2+5x)(x^2+5x+6)(x^2+5x+4)=a$$

$$\boxed{y(y+6)(y+4)=a}$$

$$x(x+1)(x+2)\cdots(x+k)=a, \quad k\in\mathbb{Z}^+$$

$$x(x+k)=x^2+kx$$

$$(x+1)(x+k-1)=x^2+kx+(k-1)=x^2+kx+i(k-i), \quad i=1$$

$$(x+2)(x+k-2)=x^2+kx+(k-2)2=x^2+kx+i(k-i), \quad i=2$$

$$(x+3)(x+k-3)=x^2+kx+3(k-3)=x^2+kx+i(k-i), \quad i=3$$

$$\vdots \qquad\qquad \vdots$$

$$\left(x+\frac{k-1}{2}\right)\left(x+\frac{k+1}{2}\right)=x^2+kx+\frac{k^2-1}{4}=x^2+kx+i(k-i), \quad i=\frac{k-1}{2}$$

$$(x^2+kx)(x^2+kx+k-1)\left[x^2+kx+2(k-2)\right]\cdots\left[x^2+kx+\frac{k^2-1}{4}\right]=a$$

$$y = x^2 + kx$$

$$y\,(y + k - 1)\,[y + 2(k-2)]\,[y + 3(k-3)] \ldots \left[y + \frac{k^2-1}{4}\right] = a$$

$$\prod_{i=1}^{k} i = 1 \times 2 \times 3 \times 4 \ldots k = k!$$

$$\prod_{i=0}^{k} [y + i(k-i)] = a \qquad \partial P = k+1 \longrightarrow \partial P = \frac{k+1}{2}$$